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1. Given that $Q \in R^{n\times d}$, $K \in R^{n\times d}$, $Z = QK^T \in R^{n\times n}$

Forward Propagation: $Z = QK^T \Rightarrow Z_{ab} = \sum_{t=1}^{d} Q_{at}K_{bt}$

Backward Propagation: $\begin{cases} \frac{\partial Z_{ab}}{\partial Q_{ij}} = \begin{cases} K_{bj}, & a=i \\ 0, & a\neq i \end{cases} \text{ and } \frac{\partial J}{\partial Q_{ij}} = \sum_{a=1}^{n}\sum_{b=1}^{n} \frac{\partial J}{\partial Z_{ab}} \frac{\partial Z_{ab}}{\partial Q_{ij}} \\ \frac{\partial Z_{ab}}{\partial K_{ij}} = \begin{cases} Q_{aj}, & b=i \\ 0, & b\neq i \end{cases} \text{ and } \frac{\partial J}{\partial K_{ij}} = \sum_{a=1}^{n}\sum_{b=1}^{n} \frac{\partial J}{\partial Z_{ab}} \frac{\partial Z_{ab}}{\partial K_{ij}} \end{cases}$

Therefore: $\begin{cases} \frac{\partial J}{\partial Q_{ij}} = \sum_{b=1}^{n} \frac{\partial J}{\partial Z_{ib}} K_{bj} \\ \frac{\partial J}{\partial K_{ij}} = \sum_{a=1}^{n} \frac{\partial J}{\partial Z_{ai}} Q_{aj} \end{cases}$

2. Let $g^{(i)} = z^{(i)} - \tilde{z}^{(i)} \Rightarrow J = \begin{cases} \frac{1}{N}\sum_{i=1}^{N} \frac{1}{2} g^{2(i)}, & |g^{(i)}| < 1 \\ \frac{1}{N}\sum_{i=1}^{N} |g^{(i)}| - \frac{1}{2}, & |g^{(i)}| \geq 1 \end{cases}$

$$\frac{\partial J}{\partial g^{(i)}} = \begin{cases} \frac{1}{N} g^{(i)}, & |g^{(i)}| < 1 \\ \frac{1}{N} \operatorname{sign}(g^{(i)}), & |g^{(i)}| \geq 1 \end{cases}$$

$$\frac{\partial g^{(i)}}{\partial z^{(i)}} = 1$$

$$\Rightarrow \frac{\partial J}{\partial z^{(i)}} = \frac{\partial J}{\partial g^{(i)}} \cdot \frac{\partial g^{(i)}}{\partial z^{(i)}} = \begin{cases} \frac{z^{(i)} - \tilde{z}^{(i)}}{N}, & |z^{(i)} - \tilde{z}^{(i)}| < 1 \\ \frac{\operatorname{sign}(z^{(i)} - \tilde{z}^{(i)})}{N}, & |z^{(i)} - \tilde{z}^{(i)}| \geq 1 \end{cases}$$